Physics-Enhanced Machine Learning

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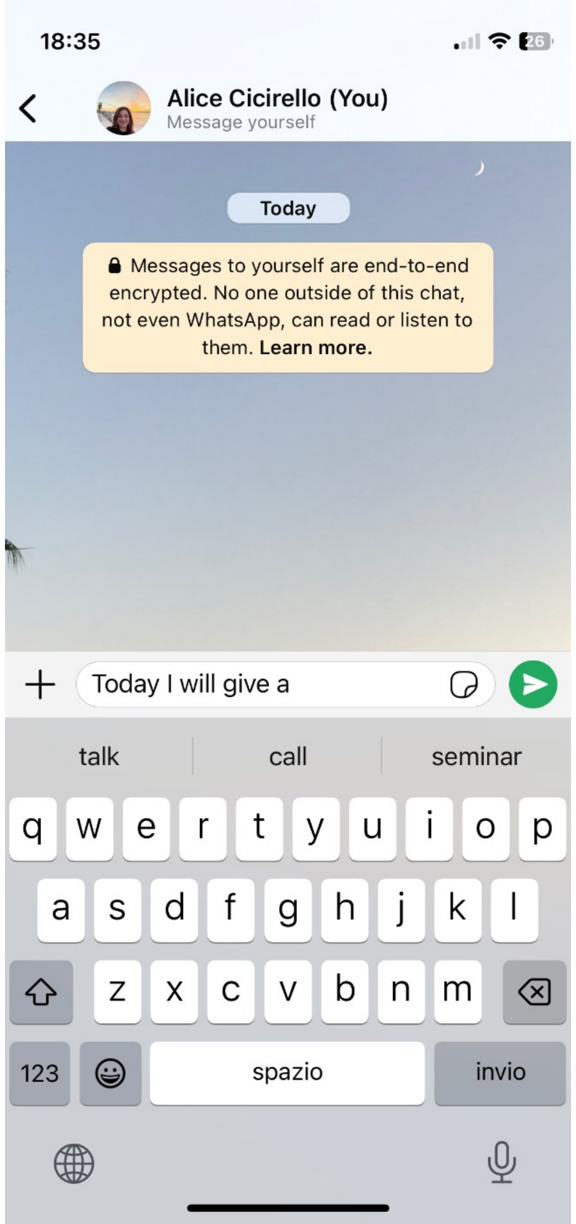
Learning objectives of this lecture

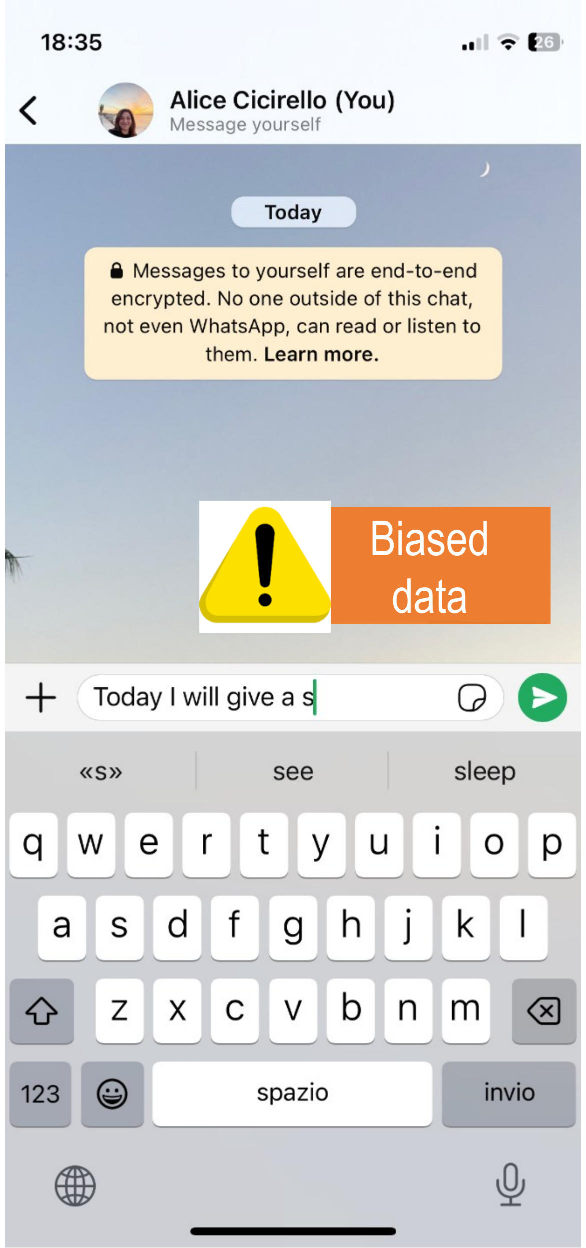
Physics-Enhanced Machine Learning

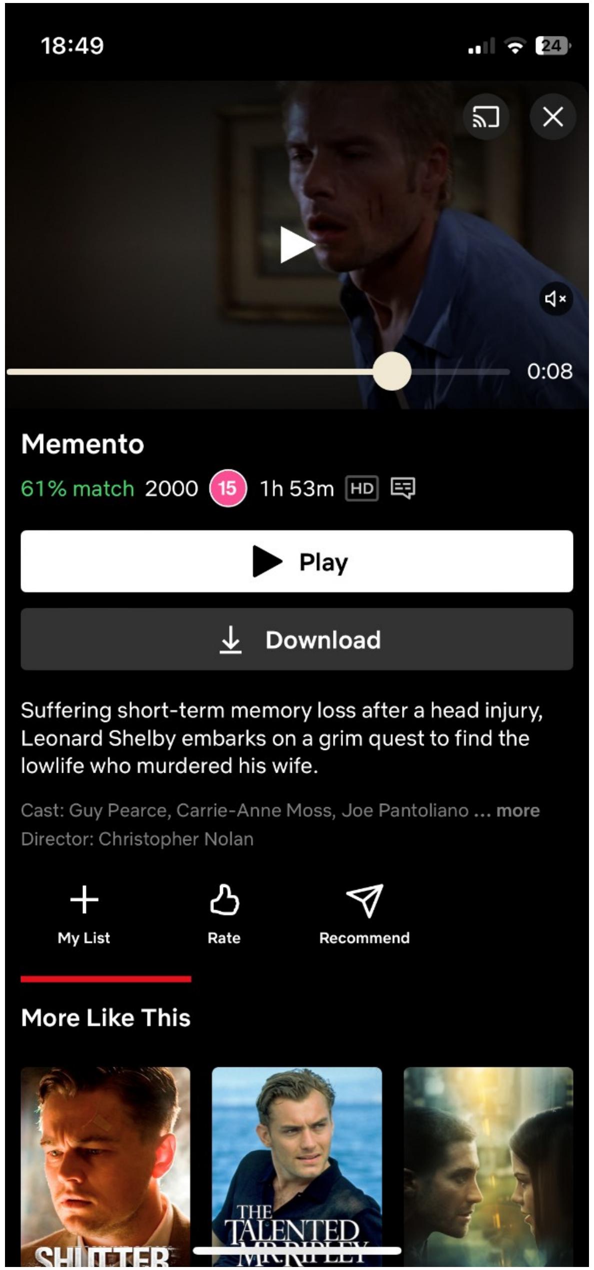
- Intro to Physics-Enhanced Machine Learning
- Overview of PEML strategies developed within the DVU group

L1: Fundamentals of Probabilistic Machine Learning (PML) → **L2: Hands-on PML** → **L3: Fundamentals of Physics-Enhanced Machine Learning (PEML)** → **L4: Hands-on PEML**

Guiding decision-making: **low risk consequences based on data**





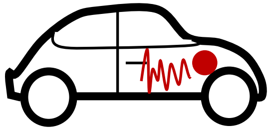


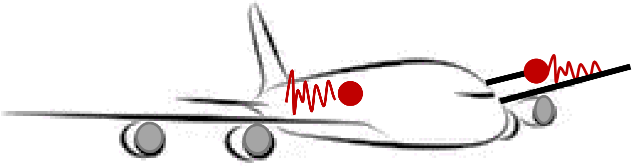
Little data

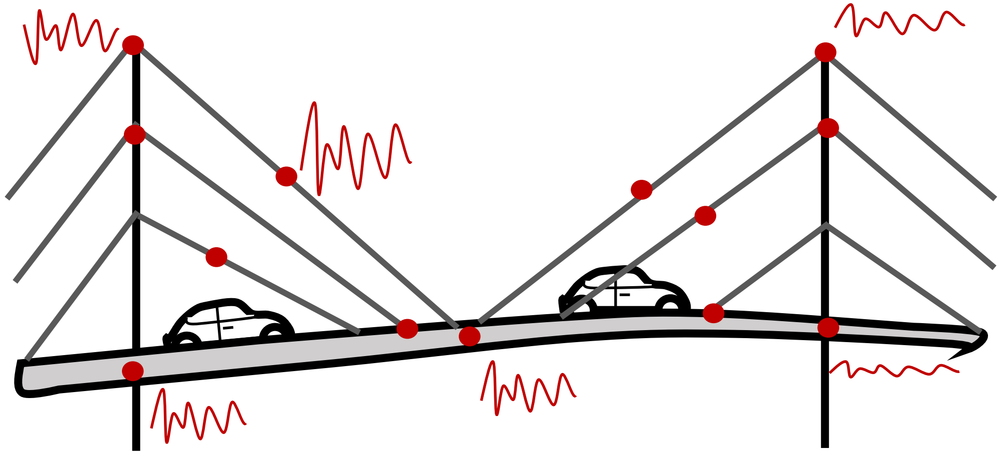
*I do not own the copyright of these pictures which are shown for educational purpose only – trained on my data

Guiding decision-making in Engineering has **high risk consequences**!

Can we just use data? What data?







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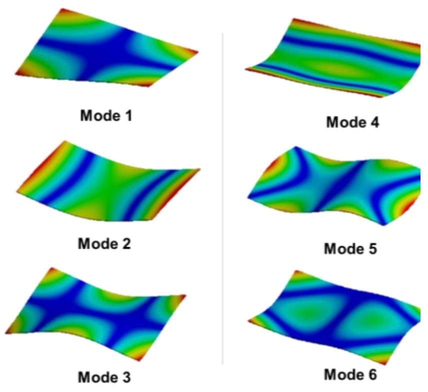
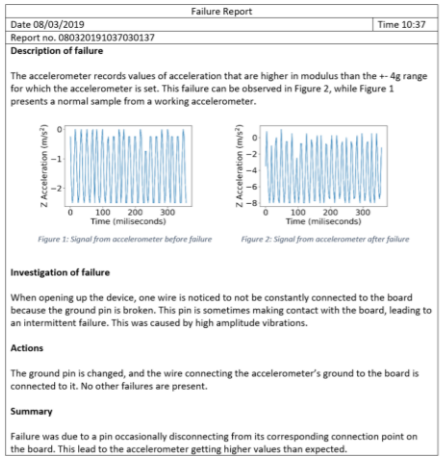
In Engineering we are dealing with **Expensive MUSIC** and **NOISE**

Data is
expensive to acquire,

Multimodal, **Uncertain**, **Sparse,**
Incomplete,
and it might be
Corrupted by confounding influences

Non-smooth and
not necessarily **Observable** phenomena**,**
Indirect, **Simulated data,**
and typically, we care about
Extreme events





Failure Report	
Date 08/03/2019	Time 10:37
Report no. 080320191037030137	

Description of failure

The accelerometer records values of acceleration that are higher in modulus than the +- 4g range for which the accelerometer is set. This failure can be observed in Figure 2, while Figure 1 presents a normal sample from a working accelerometer.

Figure 1: Signal from accelerometer before failure

Figure 2: Signal from accelerometer after failure

Investigation of failure

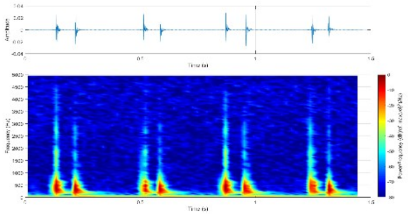
When opening up the device, one wire is noticed to not be constantly connected to the board because the ground pin is broken. This pin is sometimes making contact with the board, leading to an intermittent failure. This was caused by high amplitude vibrations.

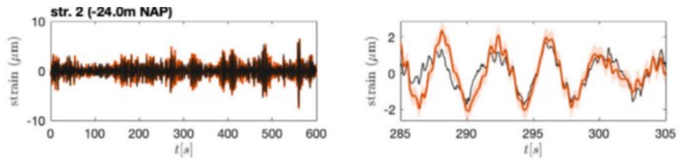
Actions

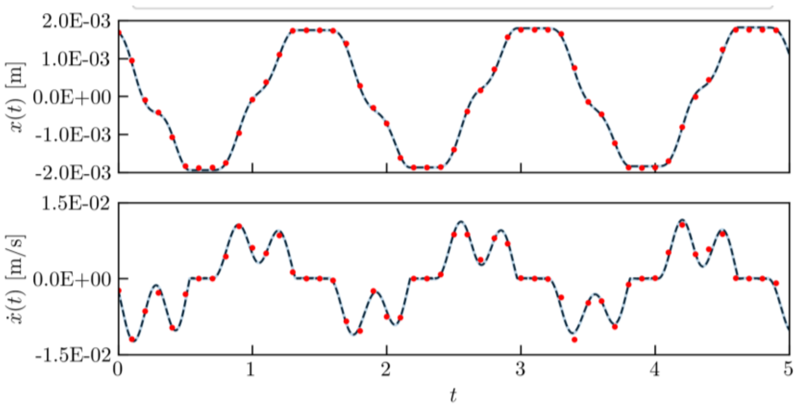
The ground pin is changed, and the wire connecting the accelerometer's ground to the board is connected to it. No other failures are present.

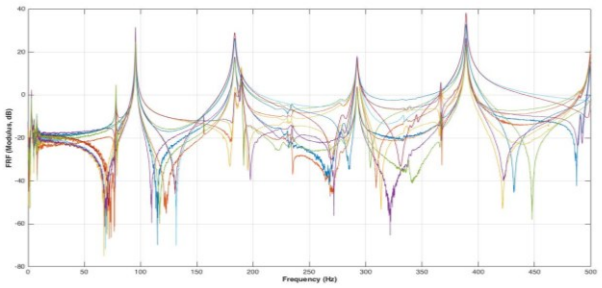
Summary

Failure was due to a pin occasionally disconnecting from its corresponding connection point on the board. This lead to the accelerometer getting higher values than expected.









Data is connected and is not enough!
Data must be transformed in relevant information!

Informative

Can E&O conditions be distinguished from health conditions?
What data? What to monitor? Where? When?

Accurate

Are the data/measurements close to the true values? (eg. low noise, calibrated, stable)

Inherent assumption made on the available data

Reliable

Would we observe the same behaviour consistently under the same circumstances over extended periods?

Machine Learning Models based on data **will be as good as the data (do we have the right data?!)**:

- Not all conditions can be observed and/or **labelled**!
- Do we have the right data for learning something about the task of interest? Is the data **up-to-date**?
- Can we trust the **quality** of the data? (e.g. are there data-biases?, was it collected in a consistent way?)
- Are there **confounding sources**?
- Do we have the **metadata**?

Tackling these challenges with **Physics-Enhanced Machine Learning (PEML)**!

A very difficult **integration of physics-data-knowledge challenge!**

Mathematical models
(multi-fidelity, multi-scale, high-dimensional, coupled,…)

Domain knowledge

Prior knowledge (observational, empirical, physical, mathematical)

First principles & appropriate biases

Computational strategies

Dealing with wrong (physics/model) biases

Small, heterogeneous, gappy, noisy data (images, time series lab test, historical data, inspection documents,…)

Multi-fidelity data (different spatial and temporal resolution; different quality)

Dealing with corrupted data

Dealing with non-informative data

Dealing with uncertainties

Providing strategies for validating solvers and prediction, forecasting and simulation models

Cicirello A., Physics-Enhanced Machine Learning: a position paper for dynamical systems investigations, **2024**

- how **much** physics vs data should be included?
- how are the physics-model, domain knowledge, and data-driven model **integrated**?

...led to **several hybrid physics-data models**!

$$f(\mathbf{x}) \approx M\left[G\left(\mathbf{x}_{model}, \boldsymbol{\theta}_p\right) + biases\right]$$

Physics – Guided ML

$$f(\mathbf{x}) \approx M\left[G\left(\mathbf{x}_{model}, \boldsymbol{\theta}_p\right) \cup D\left(\mathbf{x}_{measured}, y_{measured}, \boldsymbol{\theta}_D\right) + biases\right]$$

Physics-based model

Physics-Encoded ML

Data-driven model

$$f(\mathbf{x}) \approx M\left[D\left(\mathbf{x}_{measured}, y_{measured}, \boldsymbol{\theta}_D\right) + biases\right]$$

Physics – Informed ML

…and several other definitions:

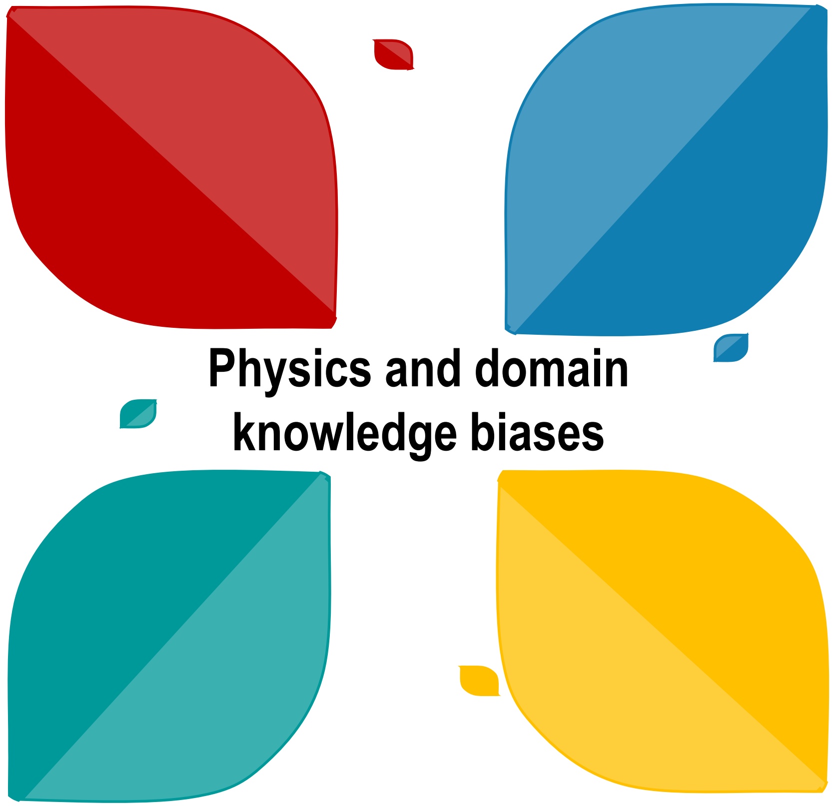
- **Grey-box models/PI spectrum:** E. J. Cross et. al., A spectrum of physics-informed Gaussian processes for regression in engineering. Data Centric Engineering, 2024
- **PEML spectrum:** M. Haywood-Alexander et. all, Discussing the Spectra of Physics-Enhanced Machine Learning via a Survey on Structural Mechanics Applications, DCE 2024
- **Scientific Machine Learning**

Cicirello A., Physics-Enhanced Machine Learning: a position paper for dynamical systems investigations, **RASD 2024**

Physics-Enhanced Machine Learning (PEML): Combining **data and “good biases”**

Observational bias:
Introducing data that embody underlying physics (e.g. lab experiments, simulations, extreme events)

Inductive bias:
Incorporate prior assumptions and physical constraints that we know of (conservation laws, symmetries, invariances, boundaries)



Learning bias:
Implement biases in an appropriate inference/learning algorithm: selection of the algorithm itself, loss functions, optimization/learning algorithm constraints

Model form/discrepancy bias:
Including “terms”, model form, and any form of partial knowledge of a physics-based model (with or without uncertainty)

- constrain **the space of admissible solutions**
- can be combined to **accelerate** training, to **reduce** computational costs and energy consumptions, and to **enhance generalisation** even with limited data
- <u>can be introduced in different parts of the algorithm (not just the loss function!)</u>

Cicirello A., Physics-Enhanced Machine Learning: a position paper for dynamical systems investigations, **RASD 2024**

(Very ambitious) Goals of Physics-Enhanced Machine Learning (PEML)

Overcoming poor generalization (to unseen conditions/new configurations) performance of ML approaches

Removing physically inconsistent or implausible predictions (e.g. improving control strategies, interaction models) with limited data

Uncertainty Quantification
physics, data, parameter, model form, model error, reducible, irreducible, … and solver/code uncertainty

Providing **explainable, interpretable, fast** and **accurate** solutions that can extrapolate to unseen conditions in the data

Learning something that we did not know before!
Informing physics: identification of unknown constitutive laws, of physics-model parameters, of governing equations, of reduced order models, of **experiments** to be carried out…

Cicirello A., Physics-Enhanced Machine Learning: a position paper for dynamical systems investigations, **RASD 2024**.:

Explainability and interpretability challenges

A definition that might work

How was a prediction made?

Interpretable

Can we understand **why** the **model** would **succeed/fail to complete a task**?

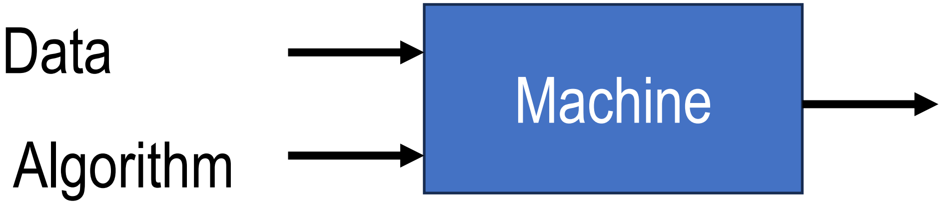
Can we communicate the predictions to support decision making?

Explainable

Cicirello A., Physics-Enhanced Machine Learning: a position paper for dynamical systems investigations, **RASD 2024**.: https://arxiv.org/abs/2405.05987

Why would a model **succeed/fail to complete a task**?

Physics-based model analysis

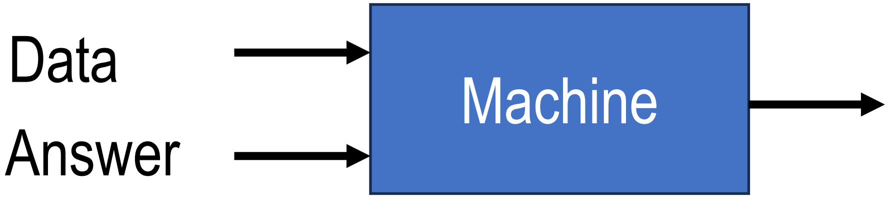


(causal model representing the physical relationship between causes and effects)

Answer (e.g.physical quantity)

- incorrect modelling assumptions (assuming a LTI system)
- incorrect physical model parameters
- Inaccurate/inappropriate solver
- …

Machine Learning



Algorithm to solve specific **tasks** (e.g. regression, classification, forecasting, learning, reduced order model,…)

- local minima, underfitted/overfitted model
- corrupted or non-informative measurements
- correlation and associational evidence that cannot be generalised
- presence of confounding influences not being properly disentangled
- Learning artefacts
- …

Cicirello A., Physics-Enhanced Machine Learning: a position paper for dynamical systems investigations, **RASD 2024**.: https://arxiv.org/abs/2405.05987

Explainability and interpretability challenges

A definition that might work

How was a prediction made?

Interpretable

Can we understand why the **model** would **succeed/fail to complete a task**?

Importance:
Humans would **trust more predictions made by an interpretable model** (not a black-box!) - **modeller**

Can we communicate the predictions to support decision making?

Explainable

Predictions must be:
- Comprehensible
- Capable of quantifying any remaining **uncertainty** in the prediction

Importance:
Critical decisions need to be presented in a format that is **understandable by humans - Decision maker**

Cicirello A., Physics-Enhanced Machine Learning: a position paper for dynamical systems investigations, **RASD 2024**.: https://arxiv.org/abs/2405.05987

Which Physics-Enhanced ML (PEML) strategy?

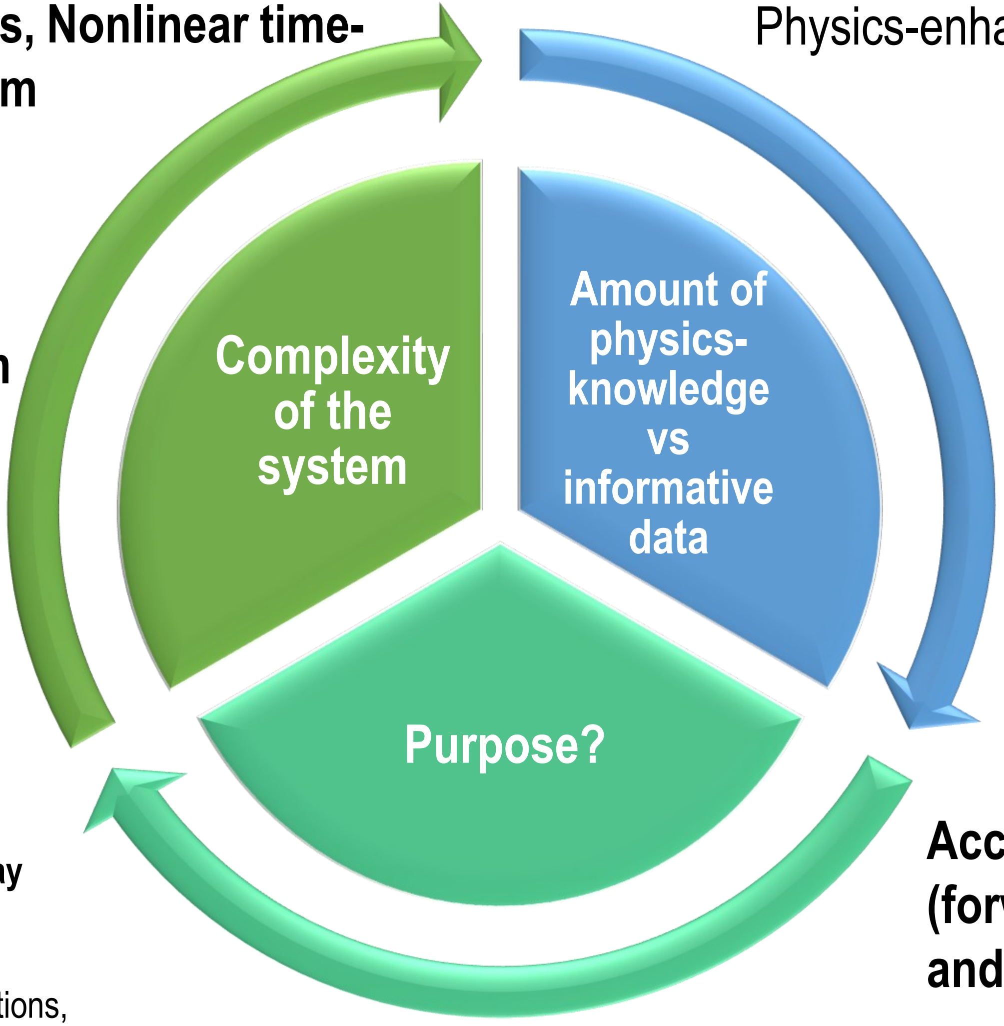
No one size fits all!

Limited to 3 examples for each category…
although there are many more!!!

Multi-scale, Multi-physics, Nonlinear time-varying dynamical system

Nonlinear Time-varying system

Linear Time Invariant System



Physics-enhanced **Probabilistic ML**

Physics-enhanced **sparse identification**

Physics-enhanced **Neural Network architecture**

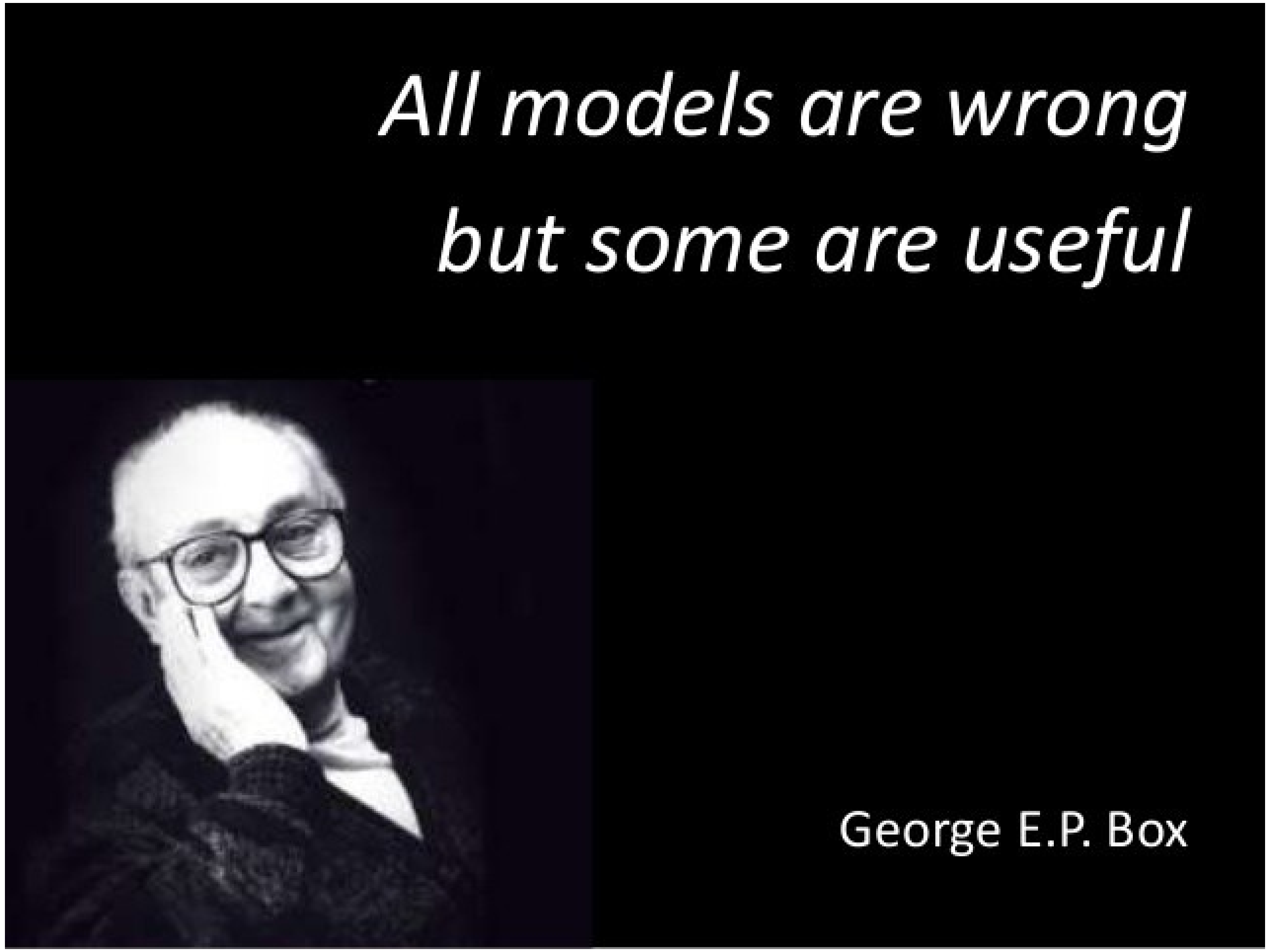
Digital Twin development

a **time-evolving** digital model based on a **two-way interaction** between the physical system (usually monitored) and the digital counterpart, that can account for uncertainty, different systems interactions, and it is used for policy and decision making.

Accelerating the solver (forward problem) and accounting for uncertainty

Identify unknown physics/terms of a model (inverse problem)

Cicirello A., Physics-Enhanced Machine Learning: a position paper for dynamical systems investigations, **RASD 2024**



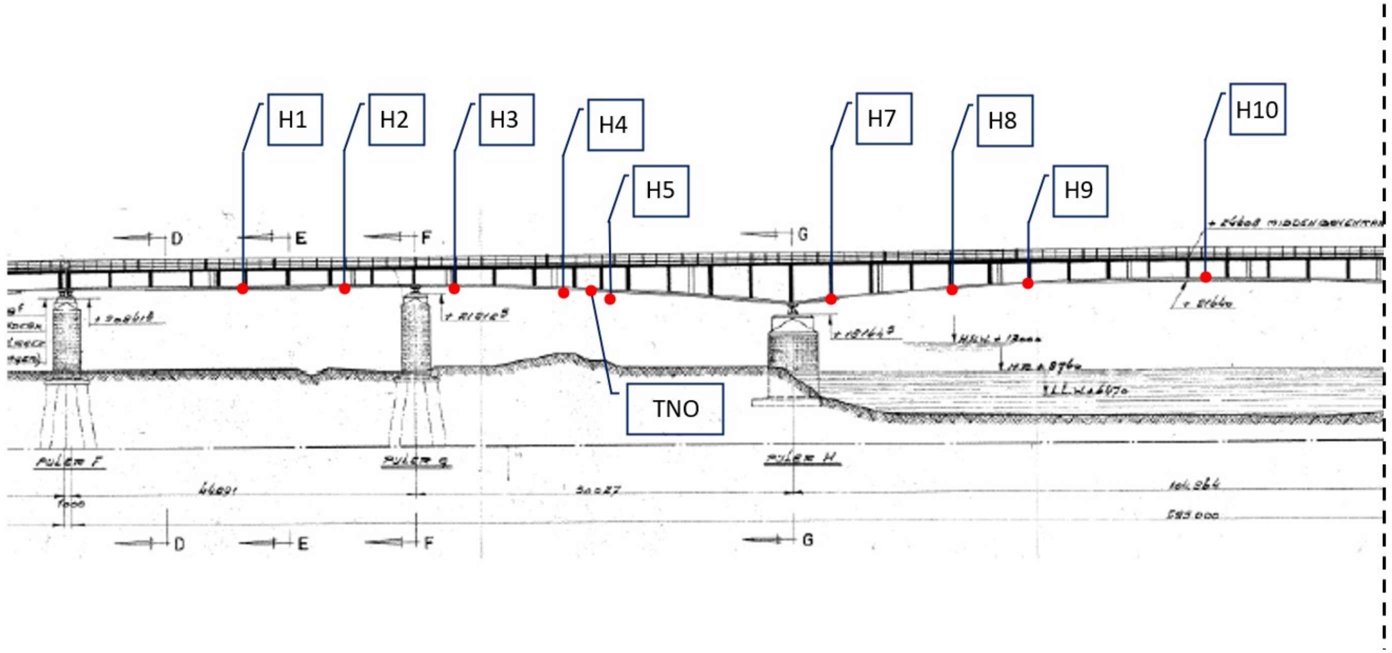
Choose wisely depending on the available information!
Don’t forget the assumptions, and wrt what dataset they have been validated!

PEML architecture for **model updating strategy**: bridges **Bayesian system identification** for structures considering **spatial and temporal correlations**

Data, Vibration, and Uncertainty Group

Physics – Guided ML

The ijsselbridge: twin girder steel roadbridge strain gauges measurements



In standard model updating strategies spatial and temporal correlations of measurements are **usually ignored**, assumed iid, leading to **wrong posterior distributions of the uncertain parameters**

Koune I., Rozsas I., Slobbe A., Cicirello A., Bayesian system identification for structures considering spatial and temporal correlation. Data Centric Engineering, 2023.

Physics-guided PEML strategy: Bayesian system identification for structures considering **spatial and temporal correlations**

Known physics-based model with uncertain parameters

Model form bias

Large number of noisy measurements

+

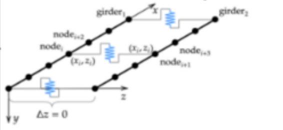
Pool of candidate models for spatial correlation

Inductive bias

Efficient calculation of loglikelihood (Multiplicative and additive model prediction error) for large datasets (10^2-10^4)

Gaussian model prediction error assumption

Physical model: $\mathbf{y}(\theta)$



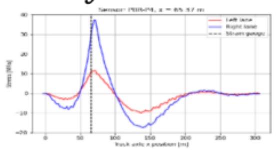
Candidate model pool:

$$\mathcal{M} = \{\mathcal{M}_1, \mathcal{M}_2, \dots\}$$

Data generating process

$$\chi \sim N(\mathbf{y}(\theta), \Sigma(\theta))$$

Data: $\bar{y}$



Likelihood:

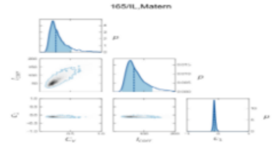
$$p(\bar{y} \mid \theta, \mathcal{M}_i)$$

Bayes rule:

$$p(\theta \mid \bar{y}, \mathcal{M}_i) = \frac{p(\bar{y} \mid \theta, \mathcal{M}_i) \cdot p(\theta \mid \mathcal{M}_i)}{p(\bar{y})}$$

Nested sampling

Posterior



Posterior predictive:

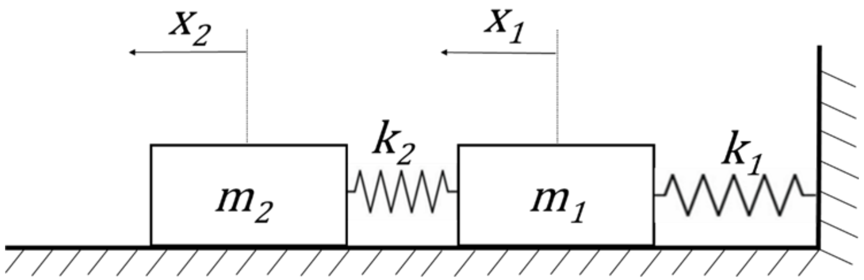
$$p(y^* \mid \bar{y}, \mathcal{M}_i) = \int_\Theta p(y^* \mid \theta, \mathcal{M}_i) \cdot p(\theta \mid \bar{y}, \mathcal{M}_i) \cdot d\theta$$

Koune I., Rozsas I., Slobbe A., Cicirello A., Bayesian system identification for structures considering spatial and temporal correlation. Data Centric Engineering, 2023.

Efficient identification of multimodal posterior distributions

Physics – Guided ML

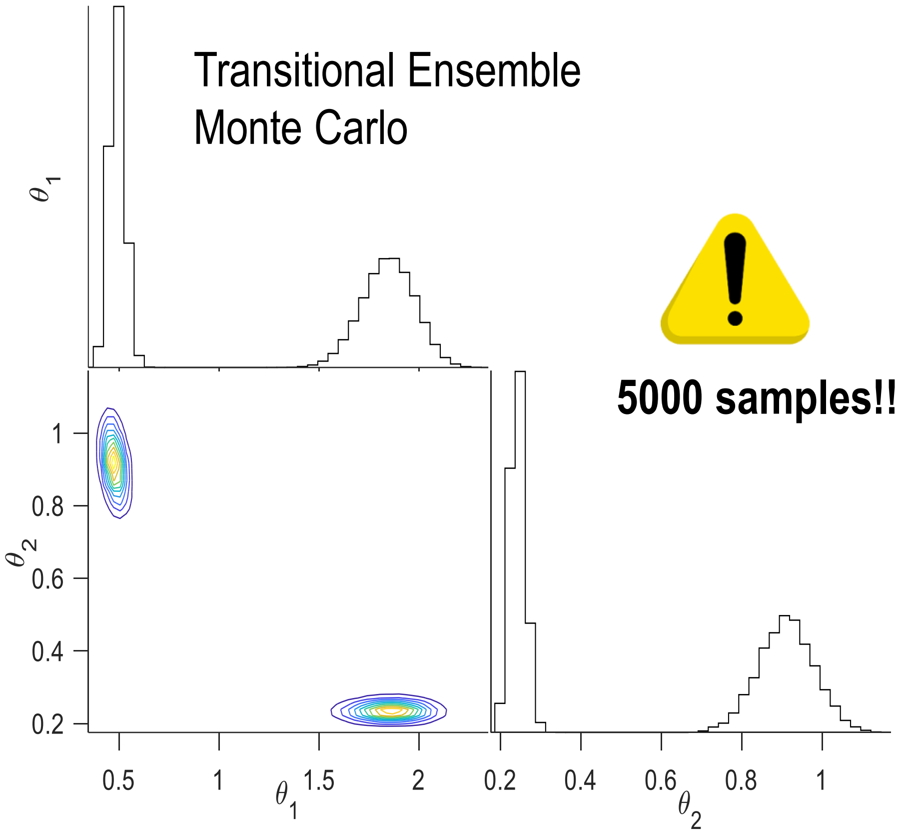
Cyclical Variational Bayes Monte Carlo



$$k_1 = \bar{k}\,\theta_1 \qquad \theta_1 \sim U(0.01, 3)$$

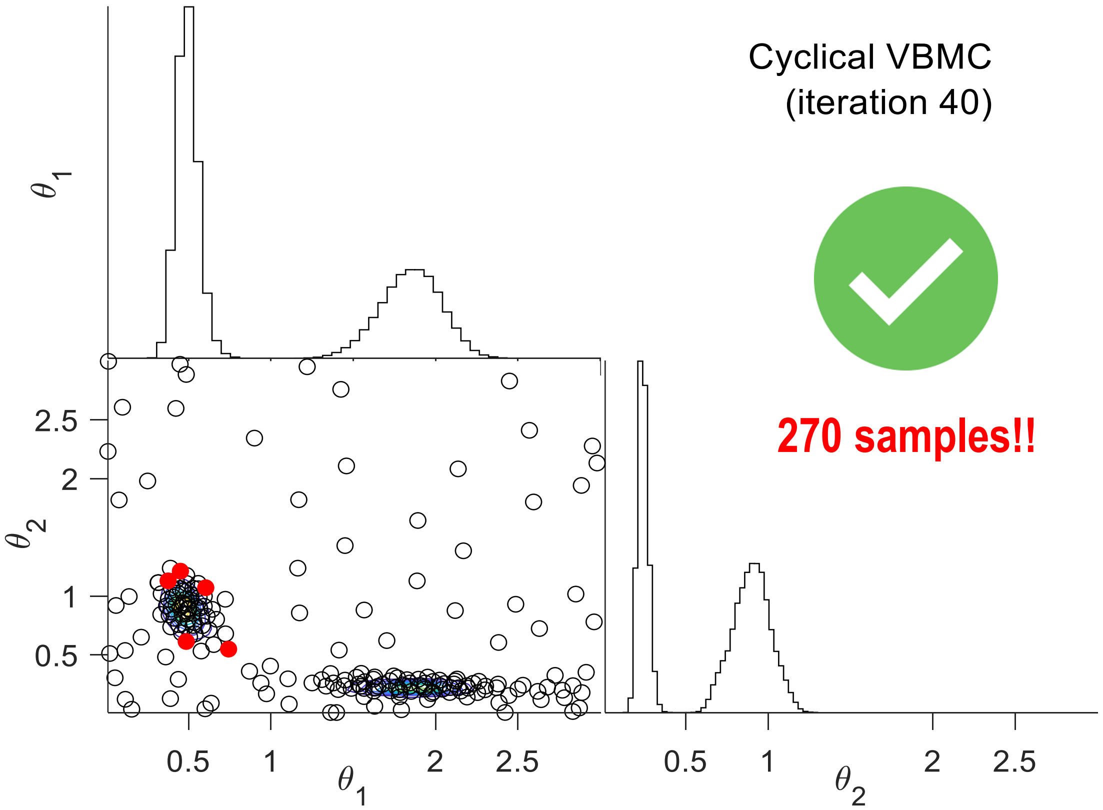
$$k_2 = \bar{k}\,\theta_2 \qquad \theta_2 \sim U(0.01, 3)$$

Transitional Ensemble Monte Carlo



5000 samples!!

Cyclical VBMC (iteration 40)



270 samples!!

Igea F., Cicirello A., Cyclical Variational Bayes Monte Carlo for Efficient Multi-Modal Posterior Distributions Evaluation, MSSP, 2023

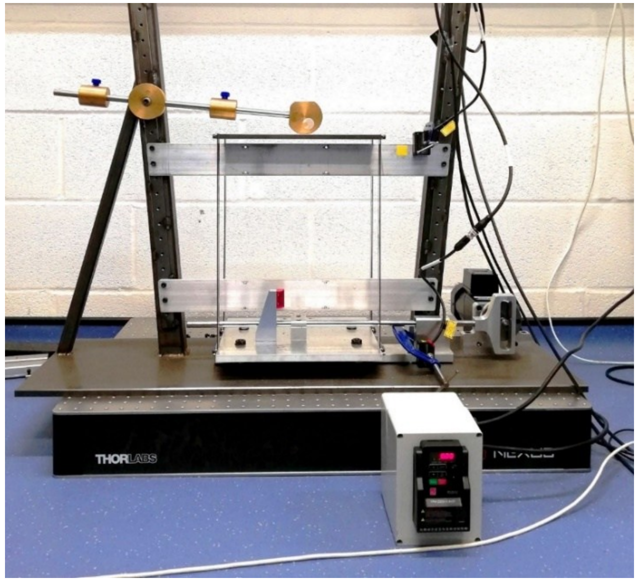
Real-world applications of PEML: lab setups

Datasets and codes available on the DVU group website!

Data, Vibration, and Uncertainty Group

Identifying governing equations and friction law or vibro-impact force **functional form**

Applied to SDOF frictional contact laboratory setup



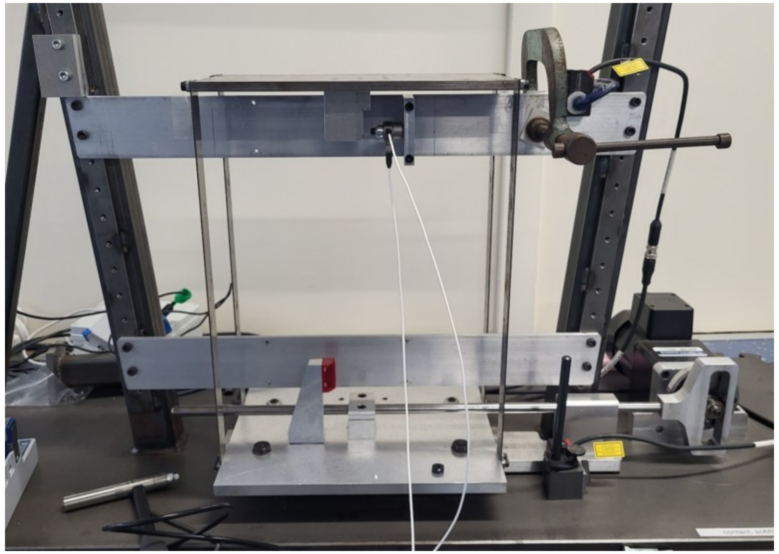
Switching **Gaussian Process** Latent force model

Marino, L., Cicirello, A., A switching Gaussian process latent force model for the identification of mechanical systems with a discontinuous nonlinearity, DCE, 2023

Physics Enhanced
Sparse Identification

Lathourakis C., Cicirello, A., Physics Enhanced Sparse Identification of Dynamical Systems with Discontinuous Nonlinearities. Nonlinear Dynamics, 2024

Applied to SDOF vibro-impact laboratory setup



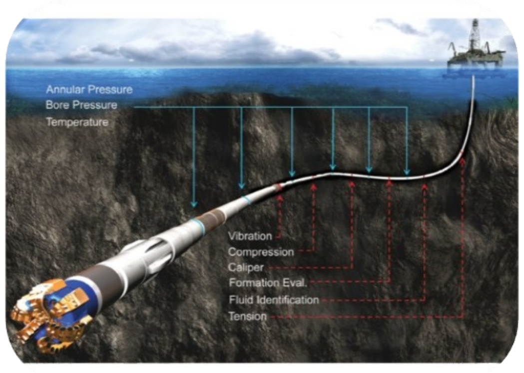
Jamie Lai

paper in the making

Physics-Encoded ML

Uncertainty in Knowledge: dealing with **non-smooth nonlinearities**

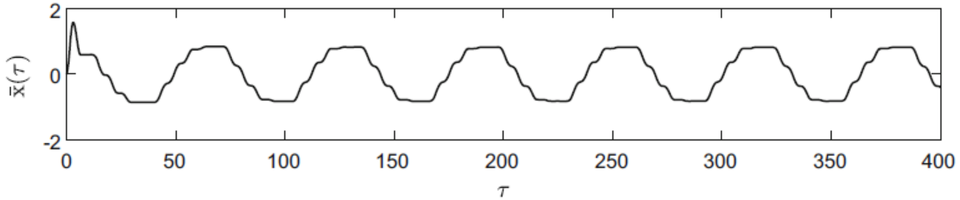
What friction **model**? What vibro-impact force function **model**?



Friction/Impact force **cannot be directly measured**

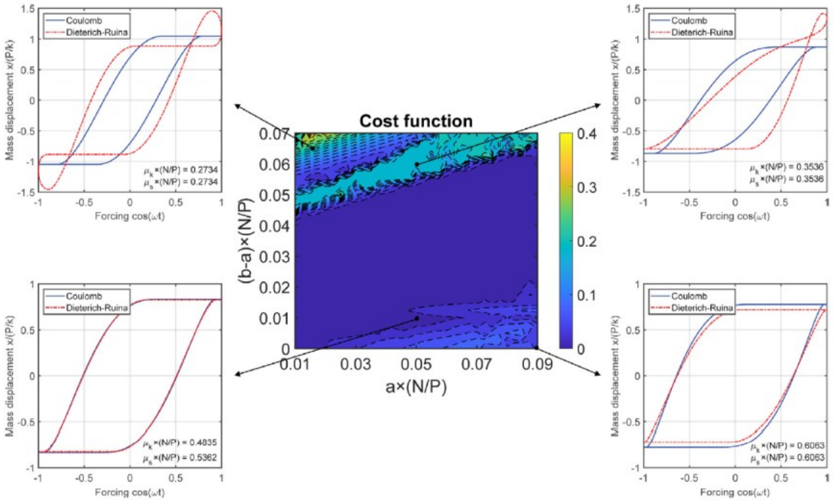
→ **Indirect** Data (sparse & Noisy)

Discontinuous and non-smooth nonlinear friction force leading to **multiple motion regimes, sharp response variations and &** <u>stiff numerical problems</u>



(a) Multiple stops stick-slip motion ($r = 0.1$, $F_\mu/F_k = 0.2$).

Two different friction laws can yield the same response! Non-uniqueness



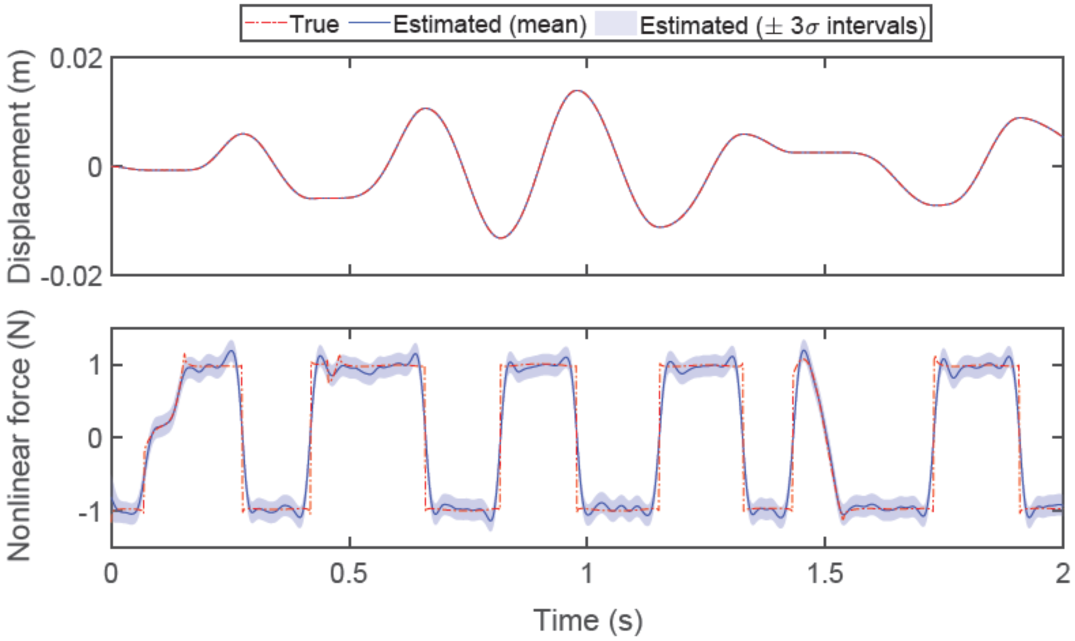
Cabboi, A., Marino, L., Cicirello, A., A comparative study between Amontons-Coulomb and Dieterich-Ruina friction laws for the cyclic response of a single degree of freedom system, European Journal of Mechanics A - Solids, 2022.

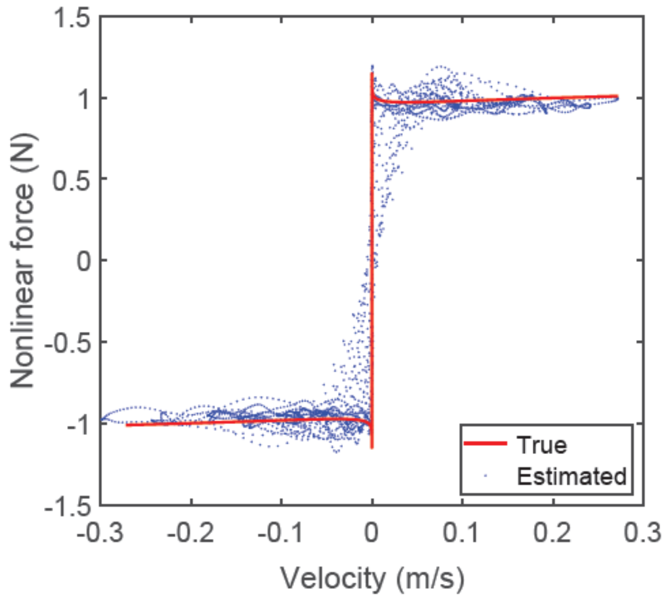
Non-smooth NSID challenge 1: Identification of **F-V curve + Force evolution and uncertainty** even during stick-slip

Known physics-based model + **Noisy dynamics response measurements** + **Input measurements** + **Gaussian Process modelling the latent friction force** = Gaussian Process Latent force model

$$m\ddot{\mathbf{z}}+c\dot{\mathbf{z}}+k\mathbf{z}+\mathbf{f}(\mathbf{z},\dot{\mathbf{z}}) = u(t),$$

$$\mathbf{f} \sim \mathcal{GP}\left(0, \kappa_i(t,t')\right)$$





- **Smoothed sharp variations**
- **Incorrect friction force description when the velocity is zero**

Rogers, T.J., Friis, T.: A latent restoring force approach to nonlinear system identification. Mech. Syst. Signal Process. (2022)

Marino, L., Cicirello, A., A switching Gaussian process latent force model for the identification of mechanical systems with a discontinuous nonlinearity, DCE, 2023

Solution to challenge 1: Switching Gaussian Process Latent force model (physics-encoded) to obtain states, F-V curve, Force evolution and uncertainty even during stick-slip

$$m\ddot{\mathbf{z}}+c\dot{\mathbf{z}}+k\mathbf{z}+\mathbf{f}(\mathbf{z},\dot{\mathbf{z}})=u(t), \qquad \mathbf{f}\sim\mathcal{GP}\left(0,\kappa_i(t,t')\right)$$

Known physics-based model

Noisy dynamics response measurements

Input measurements

+

Switching Gaussian Process to model the latent force

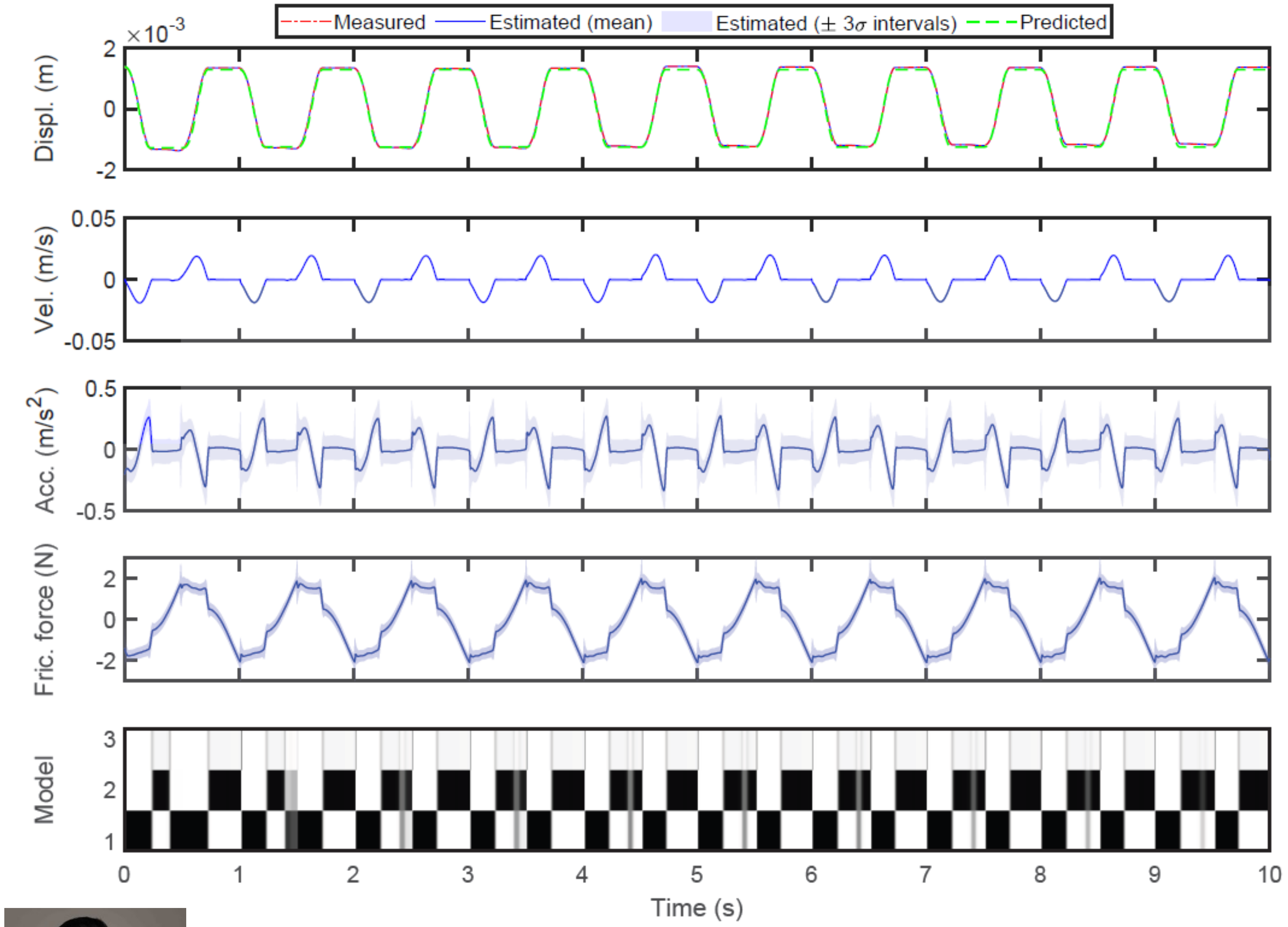
Switching models:

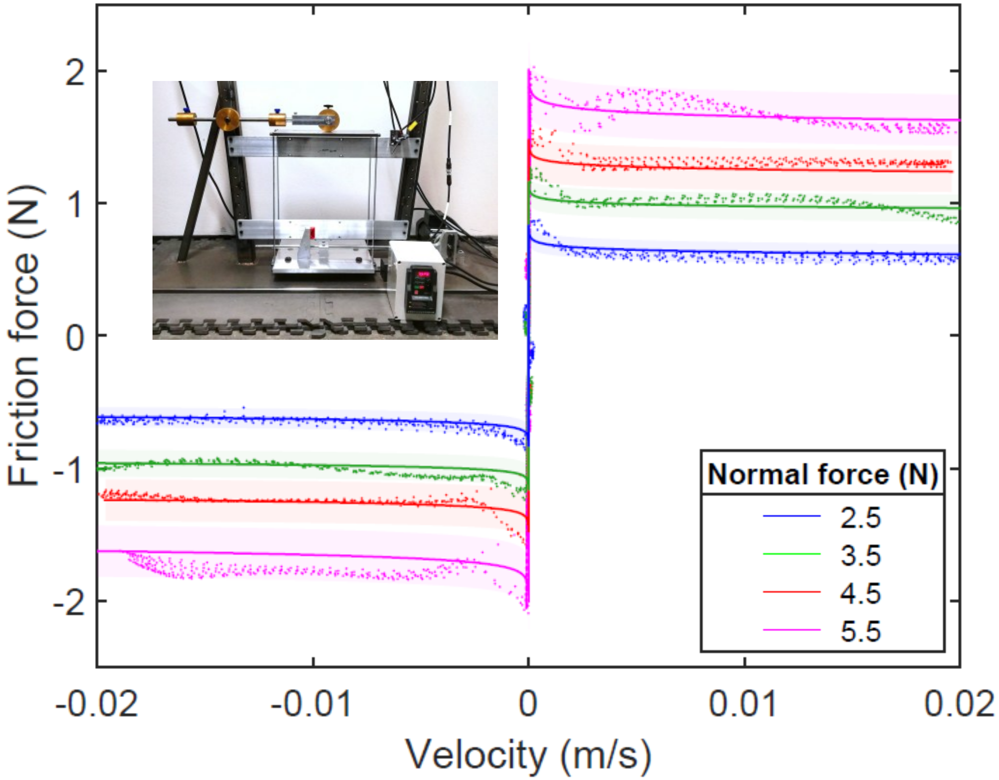
- **Sliding** response of a dry friction oscillator
- **Sticking** response of a dry friction oscillator
- Discontinuities in the latent friction force (**Resetting model**)

- Known partial model form (**discrepancy bias**);
- Specifying a set of latent force models & stick-slip constraints (**inductive bias**);
- Switching linear dynamics system governed by a discrete-time Markov model (**learning bias**)

Marino, L., Cicirello, A., A switching Gaussian process latent force model for the identification of mechanical systems with a discontinuous nonlinearity, DCE, 2023

Solution to challenge 1: Experimental validation





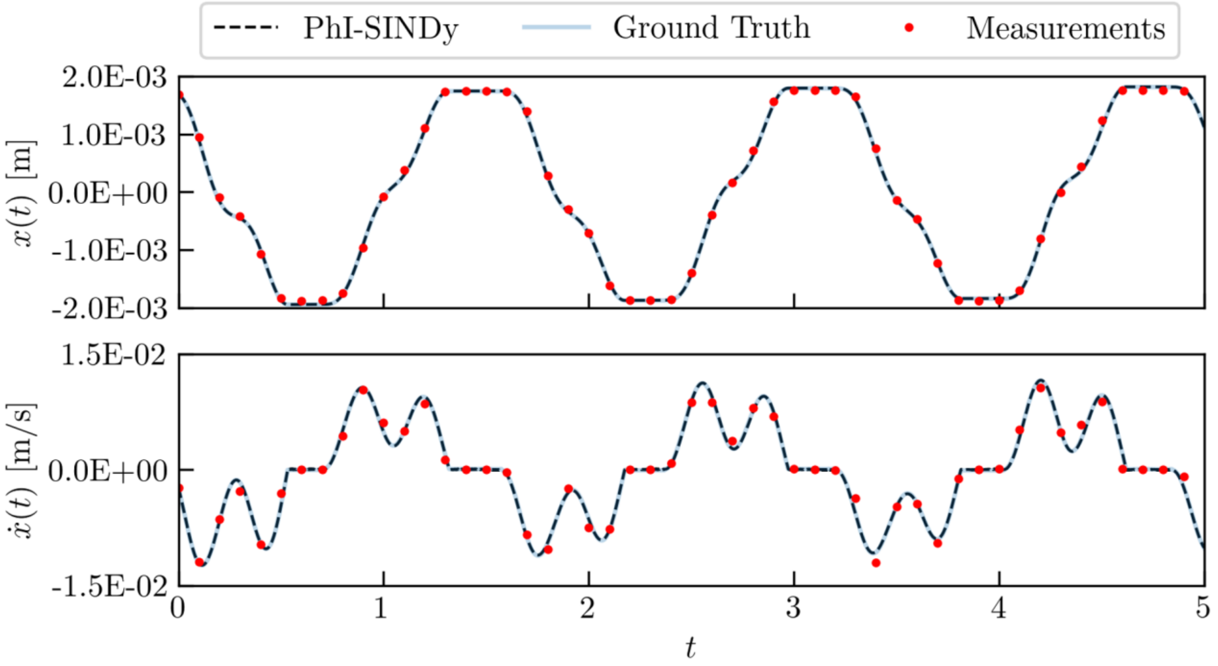
Marino, L., Cicirello, A., A switching Gaussian process latent force model for the identification of mechanical systems with a discontinuous nonlinearity, DCE, 2023

Non-smooth NSID challenge 2: identification of **the governing equations** and of the **correct friction law** with limited info → <u>generalisable model</u>

One set of (input-output) measurements + biases!

PhI-SINDy

- **Obtain Sparse governing equation**
  - Even for **high noise levels**
  - Even for **discontinuous nonlinearities**
- Applied to:
  - **MDOF systems**
  - **multiple friction nonlinearities**



Also validated with a laboratory experiment!

$$\text{Truth:}\quad \dot{y} = -0.0658\,y - 386.36\,x\ \boxed{-\ 0.0856\,\mathrm{sgn}(y)}\ + 0.0658\,\dot{u} + 386.36\,u$$

$$\text{PhI-SINDy:}\quad \dot{y} = -0.0658\,y - 386.36\,x\ \boxed{-\ 0.0855\,\mathrm{sgn}(y)}\ + 0.0658\,\dot{u} + 386.36\,u$$

Learning something we did not know before: the friction law!

Lathourakis C., Cicirello, A., Physics Enhanced Sparse Identification of Dynamical Systems with Discontinuous Nonlinearities. Nonlinear Dynamics, 2024

Our approach to uncertainty in knowledge: Physics Enhanced Sparse Identification of Dynamical Systems with Discontinuous Nonlinearities

Physics-Encoded ML

Discrepancy bias: known part of the equation

$$\frac{d}{dt}\mathbf{x}(t) = \underbrace{\mathbf{f}\left(t,\mathbf{x}(t)\right)}_{\text{Unknown functional relationship}} + \overbrace{\mathbf{g}\left(t,\mathbf{x}(t)\right)}^{\text{Discrepancy bias}}; \qquad \mathbf{x}(t=0) = \mathbf{x}_0$$

Unknown functional relationship

$$\mathbf{x}_{\text{meas.}}(t) = \left[x_1(t), x_2(t), \dots, x_N(t)\right]^{\mathrm{T}}$$

ONE SET OF Time-series noisy and sparsely collected (input-output) measurements

Learning bias: RK4, SGD, 1000 epochs, spectral method

$$f_k(\mathbf{x}(t)) \approx \mathbf{\Theta}(\mathbf{x}) \cdot \boldsymbol{\xi}_k$$

$$\mathbf{x}_{\text{pred}}(t_{j+1}) \approx \mathbf{RK4}(\mathbf{f}, \mathbf{g}, c, \mathbf{x}(t_j), t_{j+1} - t_j)$$

$$\xi_k = \min_{\Xi} \left\|\mathbf{x}_{\text{meas.}}(t) - \mathbf{x}_{\text{pred.}}(t)\right\| + \alpha \left\|\Xi\right\|_{l_1}$$

PhI-SINDy

$$\Theta(\mathbf{x}) = \left[1, x, x_{P_2}, \dots, \cos(\mathbf{x}), \dots, e^{-\mathbf{x}}, e^{-2\mathbf{x}}\right]^{\mathrm{T}}$$

Inductive bias: Library of candidate of functions (user)

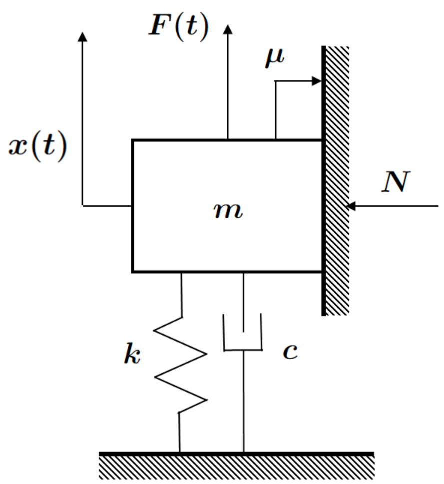
$$\frac{d}{dt}\mathbf{x}(t) = \Theta(\mathbf{x})\xi_k + \mathbf{g}(\mathbf{x}(t))$$

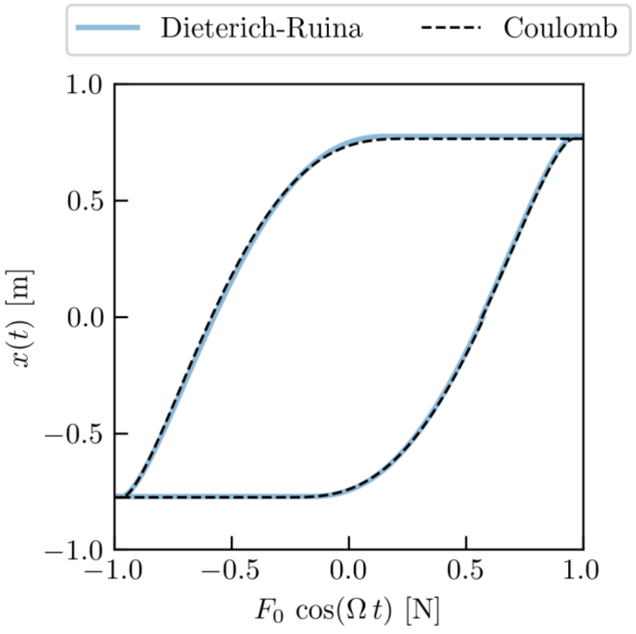
Sparse coefficients

Inductive bias: Physics constraints (user)

Lathourakis C., Cicirello, A., Physics Enhanced Sparse Identification of Dynamical Systems with Discontinuous Nonlinearities. Nonlinear Dynamics, 2024

PhI-SINDy results: distinguishing between **two friction laws** - synthetic case study





1. Displacement and velocity noisy measurements
2. Known linear part
3. Physics-constraints
4. Dictionary of candidate functions for friction force (**inductive bias**)

$$\boldsymbol{\theta}(x, y) = \left[1,\ x,\ y,\ \ln\left(\frac{|y|+\varepsilon}{V_*}\right),\ \ln\left(c + \frac{V_*}{|y|+\varepsilon}\right)\right]$$

PhI-SINDy identified the correct friction law in this inverse problem

(a)

Ground truth: $\dot{y} = -\,0.1\,y - x - 0.5\,\mathrm{sgn}(y) + \cos(0.3\,t)$

PhI-SINDy: $\dot{y} = -\,0.1\,y - x - 0.502\,\mathrm{sgn}(y) + \cos(0.3\,t)$

(b)

Ground truth: $\left[0.5 + 0.05\,\ln\left(\frac{|y|+10^{-6}}{0.003}\right) + 0.07\,\ln\left(0.022 + \frac{0.003}{|y|+10^{-6}}\right)\right]\,\mathrm{sgn}(y)$

PhI-SINDy: $\left[0.507 + 0.04\,\ln\left(\frac{|y|+10^{-6}}{0.003}\right) + 0.056\,\ln\left(0.022 + \frac{0.003}{|y|+10^{-6}}\right)\right]\,\mathrm{sgn}(y)$

Validated with lab experiments!

Known linear part

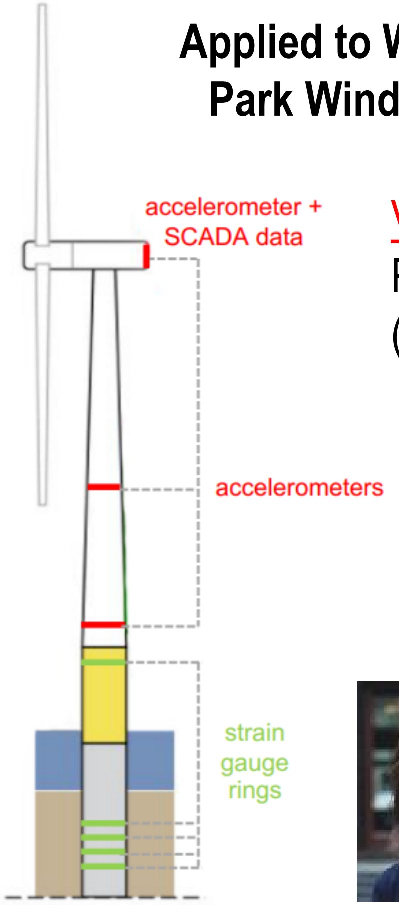
Lathourakis C., Cicirello, A., Physics Enhanced Sparse Identification of Dynamical Systems with Discontinuous Nonlinearities. Nonlinear Dynamics, 2024

Real-world applications of PEML: OWT and Ferry Quay

Datasets and codes available on the DVU group website!

Obtaining **unmeasured response states at inaccessible critical locations via virtual sensing**

Applied to Westermeerwind Park Wind Turbines data

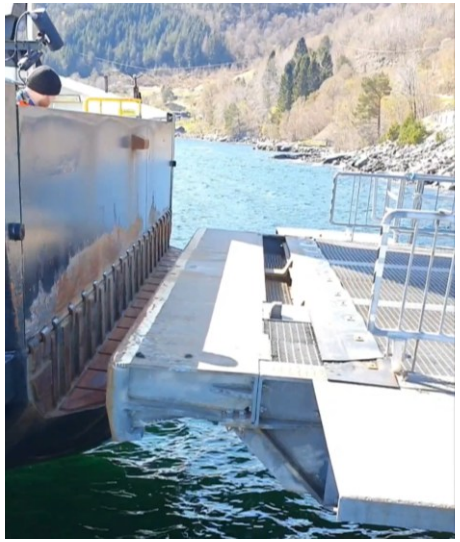


Virtual sensing via Gaussian Process Latent Force Models (**physics-encoded**)

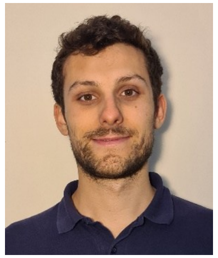
Zou J., Lourens E., Cicirello A., Virtual sensing of subsoil strain response in monopile-based offshore wind turbines via Gaussian process latent force models, MSSP, 2023.

Physics-Encoded ML

Magerholm ferry quay



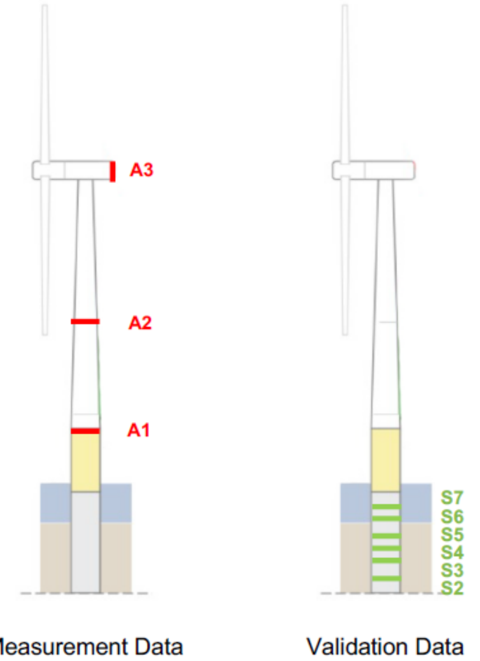
Virtual sensing and impact force estimation for a Ferry Quay (in collaboration with NTNU) via Gaussian Process Latent Force Models (**physics-encoded**)



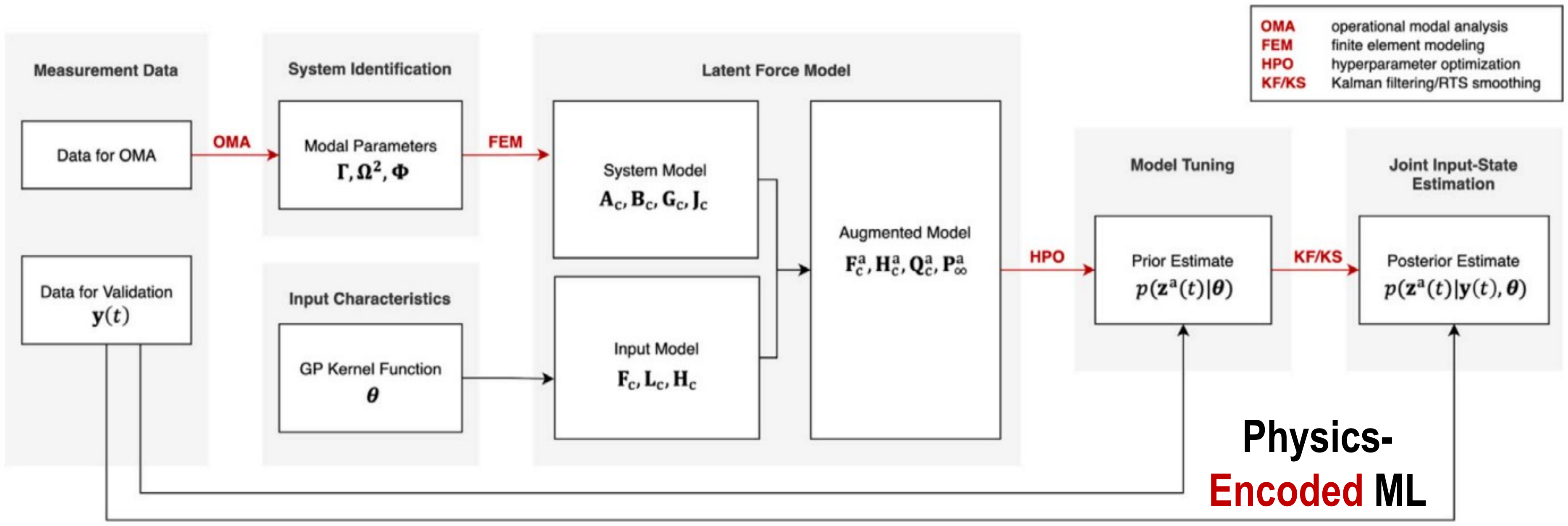
Sibille L., Nord T. S. , Cicirello A., Digital twin for virtual sensing of ferry quays via a Gaussian Process Latent Force Model. DCE, 2026

Lead by Luigi Sibille, NTNU

Obtaining **unmeasured response states at inaccessible critical locations -** Virtual sensing via Gaussian Process Latent Force Models

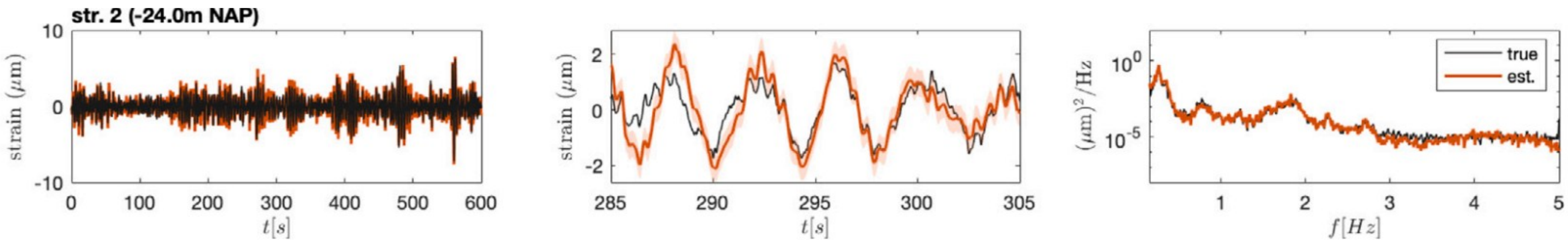


Westermeerwind Park,
The Netherlands



Physics-Encoded ML

Results at S2 during the production state

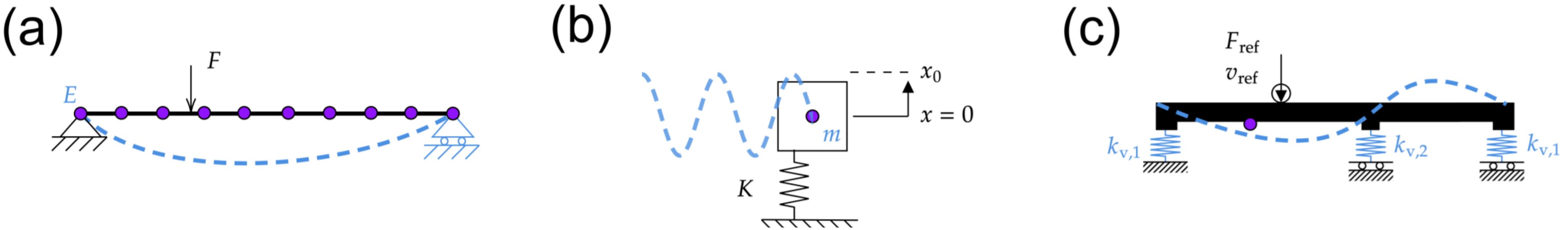


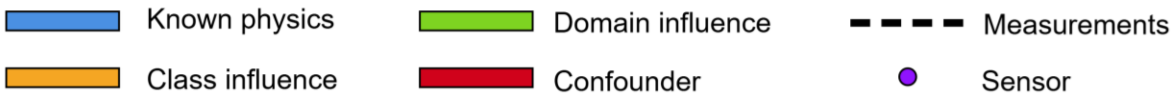
Zou J., Lourens E., Cicirello A., Virtual sensing of subsoil strain response in monopile-based offshore wind turbines via Gaussian process latent force models, Mechanical Systems and Signal Processing, 2023.

Dealing with corrupted data and partially known physics

Physics – Informed ML

- **Physics might not be complete or correct**…
- Use measurements as they are to update **the incorrect model**: yield **wrong posteriors!**
- Need to **disentangle the measurements** to correctly assess uncertainty in the latent parameters!
- **Attribute measurements contributions**: (limited) modeled physics, environmental and operational influences, presence of damage





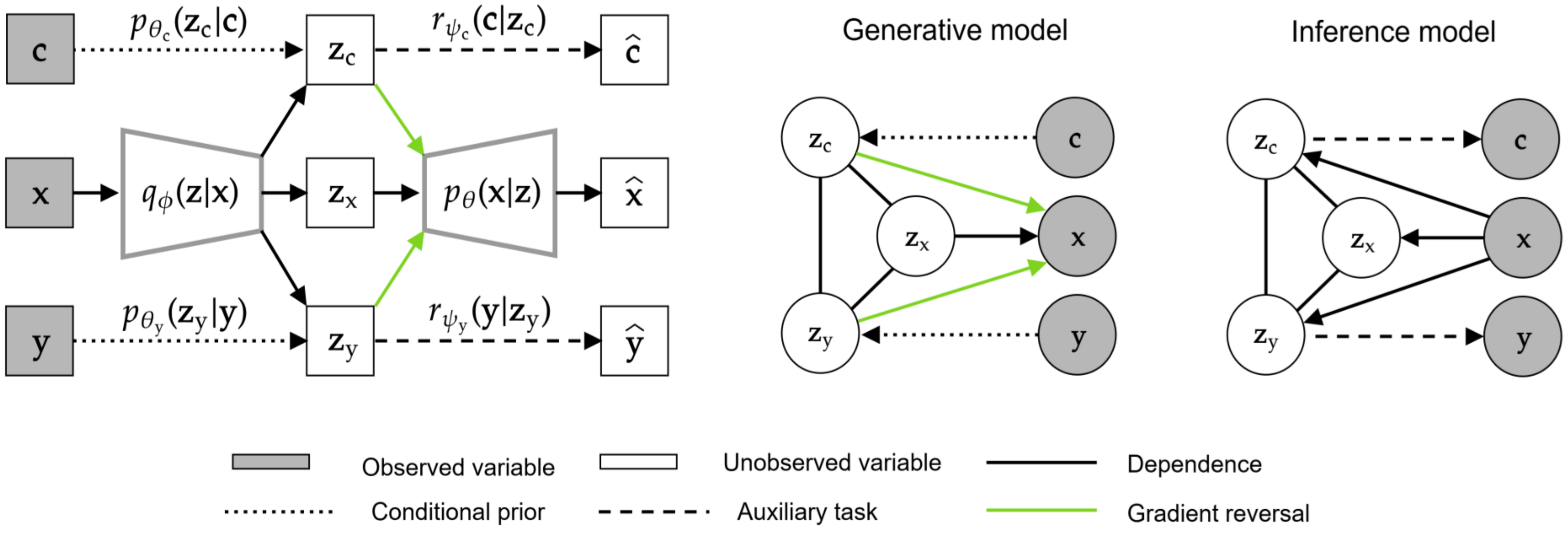
Physics-Informed Variational Autoencoder for disentanglement

Koune I., Cicirello A., Adversarial Disentanglement by Backpropagation with Physics-Informed Variational Autoencoder, 2025. DCE

Physics-Informed Variational Autoencoder for disentanglement

Physics – Informed ML

Goal: identify and **attribute variability observed in *response measurements* of** an engineering system to variability stemming from the ***modeled physics*, *domain*, and *class* influences**.



Nominal physics-based model

Dataset $\mathcal{D}=\{(x_i,y_i,c_i)\}$, composed of N triplets of

- response **measurements** x_i
- **Domain variables** c_i (measurements of environmental and operational parameters)
- **Class variables** y_i (properties of the system that are cumbersome to obtain, e.g. damage condition from inspection reports)

Important for correct **damage detection, model updating and distinction of confounding influences for critical infrastructures**

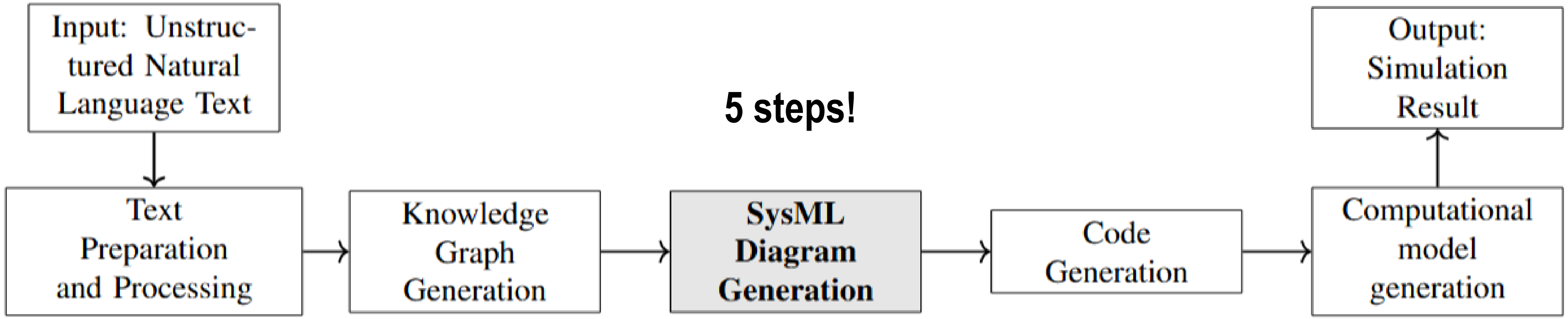
Koune I., Cicirello A., Adversarial Disentanglement by Backpropagation with Physics-Informed Variational Autoencoder, 2025. DCE:

From Text to dynamical systems models: to speeding up the design and deployment of engineering dynamical systems!

Data, Vibration, and Uncertainty Group

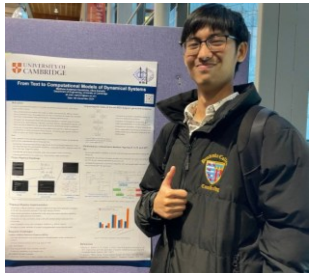
Physics – Informed ML

5 steps!



- **System modeling language diagrams (SysML)** to extract accurate information about the dependencies, attributes, and operations of components.
- **Natural Language Processing (NLP)** strategies and Large Language Models (**LLMs**) to improve intermediate outputs of the SySML diagrams automated generation
- In code generation:
  - NLP strategies are used for summarization
  - LLMs are used for validation only.
- **Domain and expert knowledge integrated** by providing a set of equation implementation templates.

Improved performance compared to results yielded by LLMs only!



Matthew Hendricks

Hendricks M.A. , Cicirello A., Text to model via SysML: Automated generation of dynamical system computational models from unstructured natural language text via enhanced System Modeling Language diagrams Arxiv: https://arxiv.org/abs/2507.06803

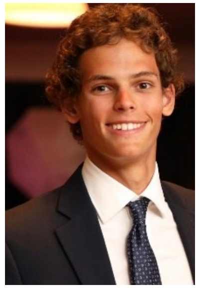
(Some of other) ongoing PEML research within the DVU group

Time-evolving systems: wind turbines

Melisa Bosaci

Will be presented @ Eurodyn

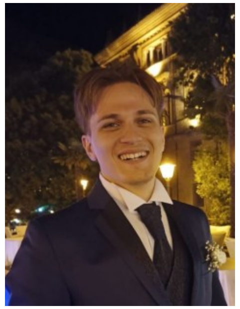
Metamaterial designs under uncertainty



Niccolò Klinger

Will be presented @ Eurodyn

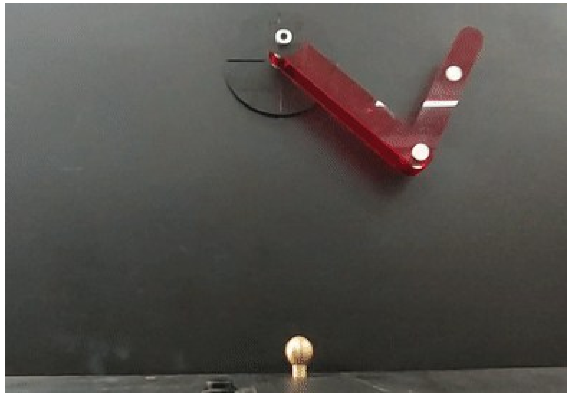
Physics-Informed Learning-based Operational Modal Analysis



Mario Visconte
PhD Student (Unical),
Visiting Researcher (UCAM)

Will be presented @ ISMA

Trajectory prediciton of chaotic systems



Rob Seabourne

Will be presented @ IAM

*I do not own the copyright of these pictures which are shown for educational purpose only

..And many more!!

…but there are still many technical challenges and opportunities

- Data coverage, resolution and correlations: spatial, temporal, frequency,..
- Non-smoothness - switching systems
- Benchmarks and validation strategies: is the model trustworthy?
- Time-varying phenomena (e.g. degradation) affecting data and models validity
- Chaotic systems
- Non-uniqueness
- Detecting and Removing Inappropriate data-bias (eg. corrupted data) – data quality
- Strategies for suggesting "extra" data/experiments/physics
- Connecting information from multi-modal data
- Instabilities, bifurcations
- Multi-physics; multi-scale
- High precision learning from small datasets
- Scalable solutions for complex problems
- Extreme, out-of-distribution events
- Learning things we did not know before: beyond fitting!
- Human-data-machine-physics-human interactions
- Where to measure, what and when? i.e. what is the useful data for a task
- Detecting & Removing Inappropriate Physics-bias (eg. wrong constitutive model)
- Accounting for noise in measurements and combined uncertainties
- Guiding decision making

Cicirello A., Physics-Enhanced Machine Learning: a position paper for dynamical systems investigations, **RASD 2024**.: https://arxiv.org/abs/2405.05987

During this lecture you learned about

- PEML strategies
- Overview of PEML strategies developed within the DVU group

group Data, Vibration, and Uncertainty Group

Next Lecture

- Hands-on PEML strategies!

Further readings

- **Deep Learning: foundations and concepts, C. Bishop, H. Bishop** (https://www.bishopbook.com/)
- **Pattern Recognition and machine learning, C. Bishop** (https://www.microsoft.com/en-us/research/uploads/prod/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf)
- **Deep Learning, I. Goodfellow, Y. Bengio, A. Courville** (https://www.deeplearningbook.org/)

I attempted to credit all the content/figures where I could find the original source. Sometimes the material is so widely used that it's difficult to be sure of the original author. Please let me know if you find missing references.